SMART INDIA HACKATHON 2026

JalRakshak

Your Water. Your Village. Your Voice.

- **Problem Statement ID -** SIH26255
- **Problem Statement Title -** Development of a solution using AI and ML to monitor the functionality of FHTC in real time at both the Gram Panchayat and household levels. This solution should provide alerts on water quality issues, allow village members to give feedback through mobile devices, and automatically sync data with IMIS/Sujal Gaon IDs.
- **Organization -** Ministry of Jal Shakti (MoJS)
- **Department -** National Jal Jeevan Mission (NJJM), Department of Drinking Water & Sanitation (DDWS)
- **Theme -** Clean & Green Technology
- **PS Category -** Hardware
- **Team ID -** 158724
- **Team Name -** Team Vistaar



Team Vistaar

THE PROBLEM & OUR SOLUTION

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THE PROBLEM

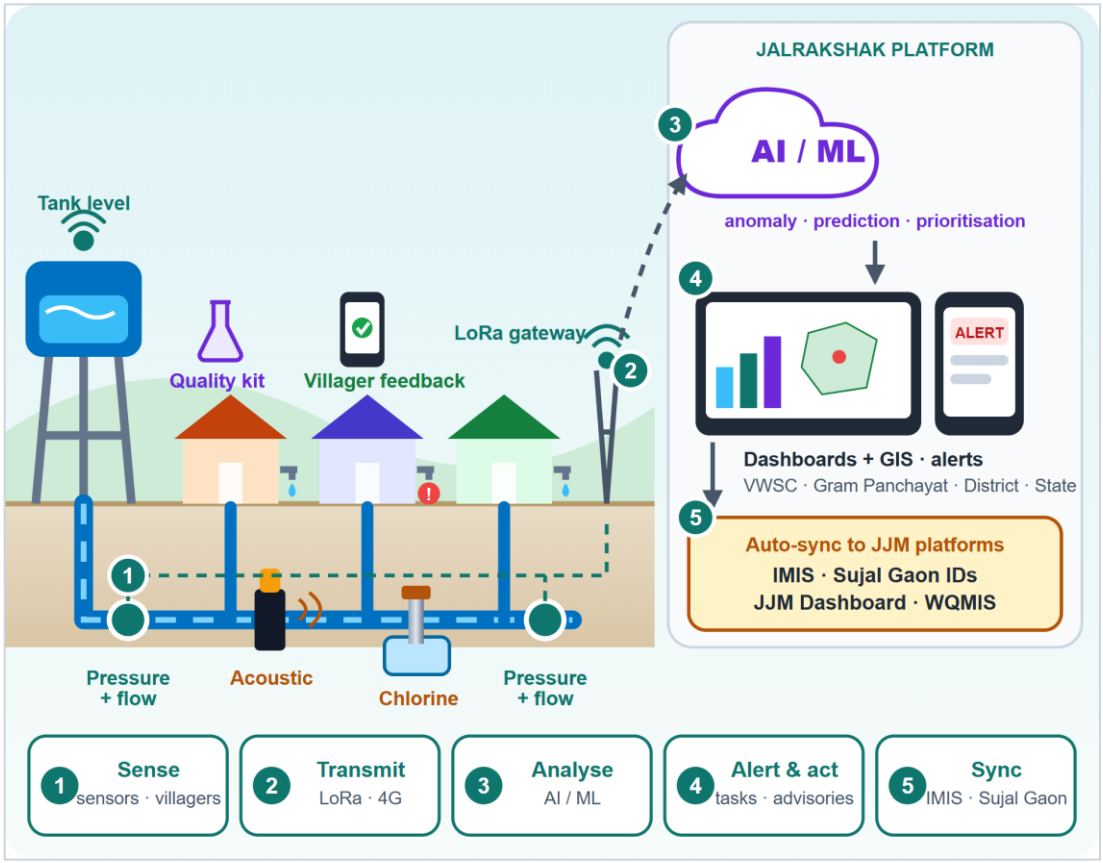
Found out too late
Problems surface through periodic reports and field inspections, by which time a tap may be dry for days

Disruptions nobody sees
Non-functional taps, intermittent supply, poor water quality, low pressure and failing pipes show up only when someone complains

Villagers have no easy voice
No quick way to report a dry tap, share a photo, or confirm that the water really came back

Data sits in separate places
Sensors, mobile apps, water-quality results and citizen feedback are rarely seen together in one village view

A tap is not a service
98.1% of surveyed homes had a tap, yet only 76.0% were assessed as functional [1]



OUR SOLUTION: JALRAKSHAK

1. **Continuous FHTC monitoring**
   sensor nodes on the pipeline + a daily villager tap check give household-level status, analysed with AI/ML
2. **Real-time alerts**
   for supply disruption, water quality deterioration, low pressure, leakages and infrastructure failure
3. **Villagers speak up**
   feedback, complaints, photographs and geo-tagged observations from a phone, in local languages
4. **Early detection and prediction**
   AI/ML spots abnormal service patterns early, predicts failures and ranks maintenance
5. **Ready to sync**
   records keyed to Sujal Gaon IDs; IMIS / JJM adapters built, official API access needed
6. **Dashboards for every level**
   GIS maps, service status indicators and analytics from VWSC to State
7. **What is new**
   Beyond sensor-only IoT pilots: HIGH alerts need sensor + villager evidence; repairs close only when readings recover and villagers confirm

SENSE Nodes + villagers → **UNDERSTAND** Rules + AI/ML → **ALERT** Alert Center → **REPORT** IMIS / JJM platforms → **FIX** Task + repair → **VERIFY** Readings recover

DEMO VIDEO (YouTube)
youtu.be/LYlQtrBbpes

For: Village Water & Sanitation Committees (VWSCs), Gram Panchayats, implementing agencies and State authorities.

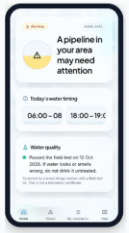
[1] DDWS, Functionality Assessment of Household Tap Connection, National Report 2024 (weighted sample estimates).

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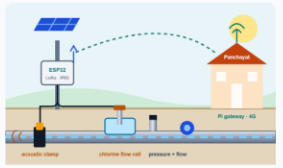
TECHNICAL APPROACH

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WHERE DATA STARTS



Villager phone



Solar sensor node

SENSE & CAPTURE

- **Sensors** — pressure · flow · acoustic · chlorine
- **ESP32 + ADS1115** — 4-20 mA loops · MicroPython firmware
- **Villager app** — React PWA · EN / TE / HI
- **Feedback** — photo · geo-tag · daily tap check

→ READINGS + REPORTS

LoRa

TRANSMIT & INGEST

- **LoRa SX1276** — 865-867 MHz, node to gateway
- **Raspberry Pi gateway** — store-and-forward buffer
- **4G · MQTT / HTTPS** — device key per node
- **FastAPI ingestion** — validation · rate limit · JWT

→ TRUSTED DATA

MQTT

AI / ML ANALYSIS

- **Rules + sensor-health gate** — offline is not "no water"
- **Isolation Forest · Theil-Sen** — anomaly + pressure forecast
- **Chlorine outlook** — repeat-fault history
- **DBSCAN + evidence fusion** — complaint clusters · tap status

→ TAP STATUS + RISK RANK

REST

ALERT, ACT & SYNC

- **Alert Center** — acknowledge · escalate
- **Maintenance planner** — task · repair · verified restoration
- **Dashboards + GIS** — VWSC → District → State
- **Auto-sync** — IMIS · Sujal Gaon IDs · JJM · WQMIS

→ ACTION + REPORTING

TECHNOLOGY STACK

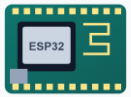
- **APP** — React · Vite · Tailwind; MapLibre · i18next (EN/TE/HI)
- **BACKEND** — Python · FastAPI; SQLAlchemy · JWT
- **DATABASE** — PostgreSQL + PostGIS; SQLite for development
- **AI / ML** — scikit-learn · NumPy; Isolation Forest · DBSCAN · Theil-Sen
- **FIELD / IoT** — ESP32 MicroPython · ADS1115; LoRa SX1276 · Pi · MQTT
- **INTERFACES** — REST + OpenAPI; GeoJSON · CSV
- **DEPLOYMENT** — Docker · Nginx; solar node · IP65 enclosure

BACKEND AND DATA

POSTGRESQL + POSTGIS primary data platform
Auth (JWT, role-based by village) · photo store (GPS stripped) · audit log · reading retention

GATEWAY BUFFER fallback
keeps readings while 4G is down; replays in order

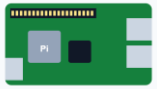
Proposed hardware and how it connects (firmware tested in simulation; bench build next; circuit on the next slide)



ESP32 node
Reads sensors through ADC, I2C and GPIO; sleeps between samples



LoRa SX1276
SPI radio, 865-867 MHz, node to village gateway



Pi gateway
Buffers readings, sends over 4G MQTT to the ingestion API



Acoustic sensor
Piezo clamp at critical points; leak-noise indicator, confirmed by flow and pressure



Chlorine sensor
4-20 mA probe in a flow cell at the tail end; measured mg/L, low-chlorine alert



Pressure x2
4-20 mA transmitters, head and tail; low-pressure alert



Flow meter x2
Pulse output, head and tail; a gap above its normal baseline flags a possible leak



Ultrasonic level
JSN-SR04T on the tank; supply and tank health

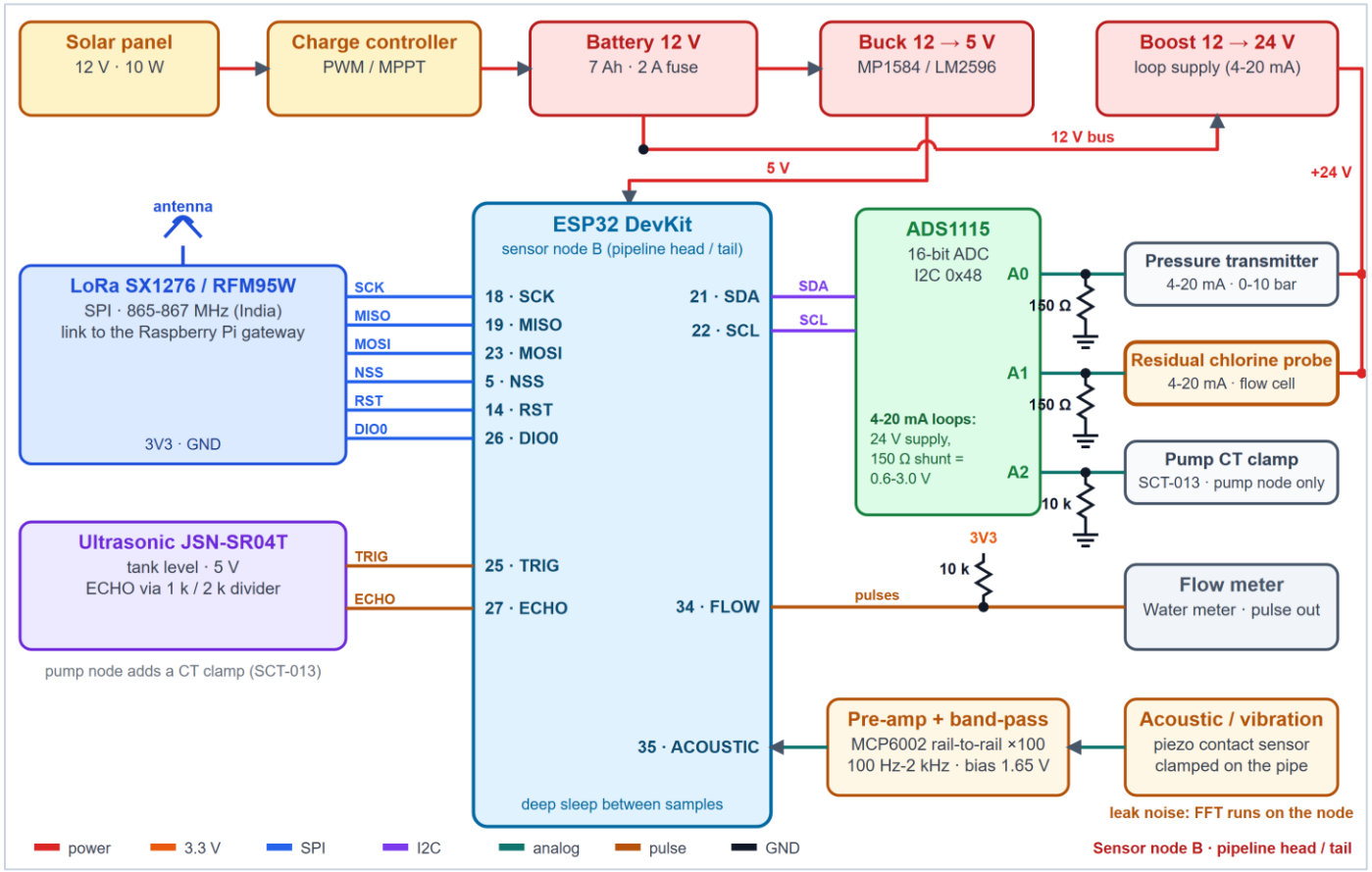


Solar + battery
12 V 10 W panel, 7 Ah battery, IP65 box; buck and boost rails

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FEASIBILITY AND VIABILITY

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Circuit of sensor node B (pipeline head / tail): ESP32 with LoRa, ADS1115, 4-20 mA loops, flow, acoustic and tank-level inputs.

Scalability

- One village, then a cluster per gateway, block, district, state
- Stateless APIs; PostgreSQL + PostGIS
- Same node design in every village

Interoperability

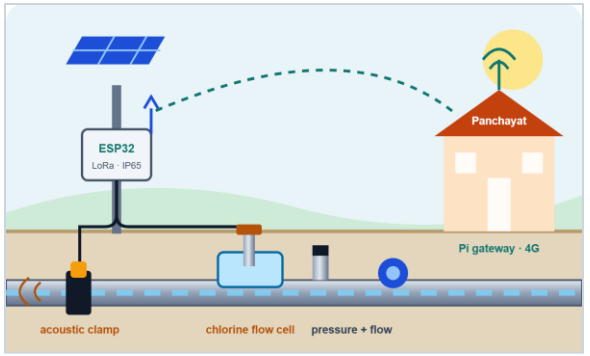
- Keyed to Sujal Gaon and Sujalam Bharat IDs
- Sync adapters built and mock-tested; live IMIS / JJM / WQMIS sync needs official API access
- Open formats: REST/OpenAPI, GeoJSON, CSV

Data security

- Role-based access by village; hashed passwords
- Photo location data stripped; location shared separately by opt-in
- No personal data sent to other platforms

Ease of deployment

- Solar node, plug-in LoRa gateway, no grid cabling
- Mobile web app: nothing to install
- Docker deployment, one configuration file



Node installation concept

- Solar-powered pole node with an IP65 box
- Clamp-on acoustic sensor: no digging
- Chlorine flow cell at the tail end
- LoRa to the Panchayat gateway

Feasibility analysis

- Off-the-shelf parts: ESP32, ADS1115, LoRa, Raspberry Pi
- Platform on demo data; 278 tests incl. firmware on a simulated board
- AI/ML runs on demo data now; to be validated on pilot data

Potential challenges and strategies to overcome them

False alarms: persistence windows; HIGH needs two kinds of evidence

Sensor fault or drift: health gate; chlorine probe checked with a DPD kit

Rural network: gateway buffers readings, sends when online

Trust in AI: every score shows its reasons

Leak vs noise: per-pipe acoustic baseline plus flow gap

Platform access: adapters ready; official access and specifications needed

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IMPACT AND BENEFITS

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USERS INVOLVED

Villagers
Report problems with a photo, check their tap daily, and get alerts and advisories in their own language

VWSC and Gram Panchayat
See alerts, household tap status and a ranked repair list; confirm that water is really restored

Technicians and operators
Receive tasks with the pipeline, evidence and location; close them when readings recover

Implementing agencies
Plan preventive maintenance from risk scores and fault history instead of waiting for breakdowns

District and State authorities
Multi-level analytics and GIS; data prepared for IMIS, Sujal Gaon IDs and the JJM platforms

SOCIAL BENEFITS

- Dry taps get fixed sooner, and the villager is told when water is restored
- A voice for every household, in local languages
- Boil-water advisory to homes when a field test fails, plus a confirmatory lab test request
- Tail-end homes stop being invisible

ECONOMIC BENEFITS

- Inspect the riskiest pipes first: fewer repeat repairs and wasted visits
- Leaks near monitored points are flagged earlier
- Less treated water lost to unseen leaks
- Evidence replaces guesswork in planning and budgets

ENVIRONMENTAL BENEFITS

- Early leak detection saves treated water and pumping energy
- Solar-powered nodes: no grid cabling
- Low-power LoRa links and sleep cycles
- Measured chlorine helps avoid both under- and over-dosing

EARLY DETECTION AND PREDICTION, THEN ACTION

DETECT EARLY	PREDICT	ACT BEFORE IT SPREADS
Acoustic leak noise, pressure trend, chlorine probe and sensor health, before villagers notice	Pressure and chlorine trends; pipes ranked by fault evidence, with an evidence score and what is uncertain	Alert, planned inspection, repair, and restoration verified from readings

WHAT A PILOT WILL MEASURE

1. Time from fault to alert, vs complaint-led detection
2. Time from alert to verified restoration
3. Share of days each household reports water
4. False-alarm rate under the two-evidence rule
5. Low-chlorine events caught before a kit test
6. Repairs planned from the ranked risk list

FUTURE SCOPE **Public village water report card** (share of working taps, repair time, last quality test; no personal data) **WhatsApp / SMS task alerts** to linemen and pump operators

ROADMAP Idea submission → Working software platform → Bench rig with real nodes → Finale demo → Field pilot

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RESEARCH AND REFERENCES

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Details / links of the reference and research work

[1] DDWS, Functionality Assessment of Household Tap Connection, National Report 2024
jaljeevanmission.gov.in/sites/default/files/2025-12/FHTC_Na...

[2] PIB, 13 Aug 2026: 75% of rural habitations covered with Sujal Gaon IDs
pib.gov.in/PressReleasePage.aspx?PRID=2298959

[3] PIB, 30 Dec 2025: Jal Seva Aankalan launched
pib.gov.in/PressReleasePage.aspx?PRID=2209798

[4] Cabinet, 10 Mar 2026: JJM extended to Dec 2028, service-delivery focus
pmindia.gov.in/en/news_updates/cabinet-approves-extension-of-jal-jeevan-mission...

[5] JJM Dashboard (coverage, GIS, IoT pilot view)
ejalshakti.gov.in/JJMReport/JJMIndia.aspx

[6] JJM-WQMIS (laboratory and FTK water-quality testing)
ejalshakti.gov.in/WQMIS/

[7] PIB, 31 Mar 2021: JJM deploys sensor-based IoT devices (off-grid pilot)
pib.gov.in/PressReleaseIframePage.aspx?PRID=1708701

[8] Jal Jeevan Samvad, Oct 2023: IoT water-quality monitoring with complaint tracking
jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeev...

[9] Systematic review: ML for anomaly detection in smart water metering networks (2024)
sciencedirect.com/science/article/pii/S1474706524000160

[10] SIH 2026 problem statement SIH26255, Ministry of Jal Shakti
sih.gov.in/sih2026PS

Proposed sensor-node hardware (not yet assembled)


ESP32


LoRa


Pi gateway


Acoustic


Chlorine


Pressure


Flow

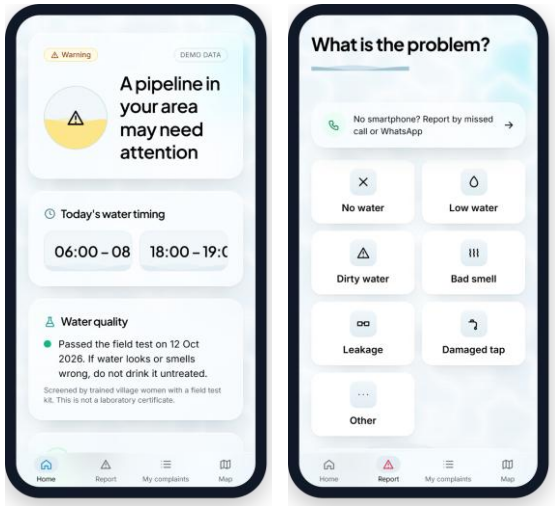

Tank level


Solar

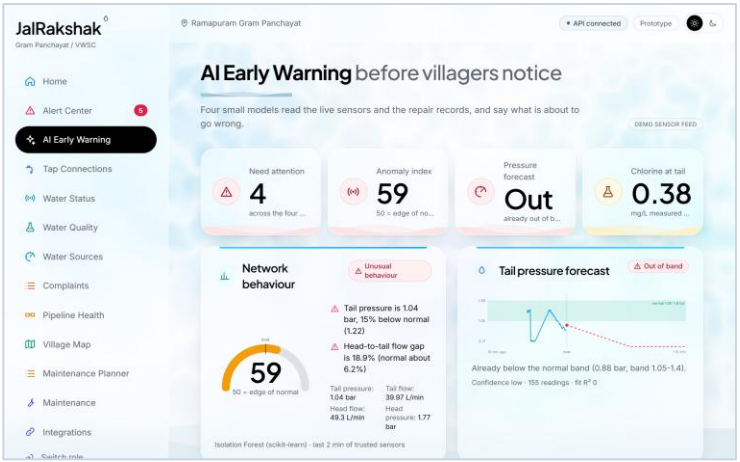
ESP32-DevKitC + ADS1115, RFM95W LoRa, 4-20 mA pressure and chlorine probes, pulse water meter, piezo clamp, JSN-SR04T. Vendor photos of single parts, not a built prototype; bench build next.

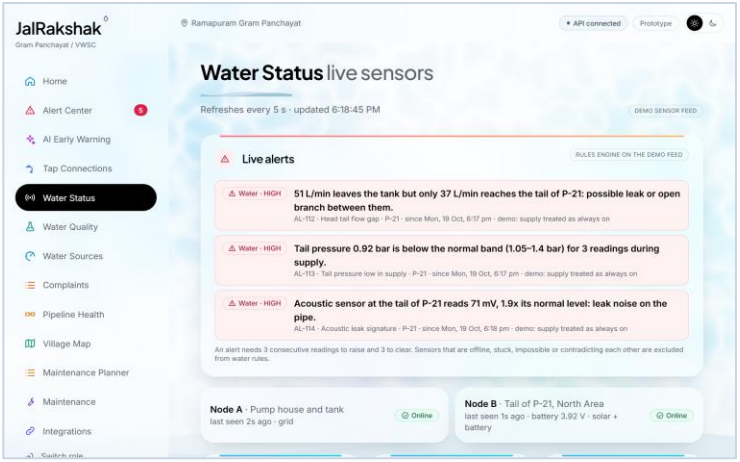
pH, TDS and turbidity come from field-test-kit entries: screening indicators that trigger confirmatory laboratory testing, not a safety certification.

Working platform screens



Villager app: tap status and water quality (left); report a problem with photo and opt-in location (right)





Gram Panchayat: AI Early Warning with measured chlorine (top right) and Water Status with acoustic leak alerts (bottom right).

Demo video: youtu.be/LYlQtrBbpes

Source code: github.com/Thrisha236/JalRakshak (platform, node firmware, gateway, circuit and architecture files)